

Yijian Lu

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EDUCATION

The University of Hong Kong (*QS World University Rankings 2026: 11th globally*)

HongKong, China

MSc(Eng) in Industrial Engineering and Logistics Management

Sep 2025 - Nov 2026

Expected: Distinction, Frontiers in Robotics and Intelligent Systems (A-); Digital Enterprises and E-commerce (A-); Frontiers in Data and Systems Engineering (B+); Machine learning & applications (B+)

Nanjing Agricultural University (*Project 211 and Double First-Class University*)

NanJing, China

BEng in Logistics Engineering

Sep 2021 - Jun 2025

Average Score: 88/100, Equipment of Logistics (96); Planning and Design of Distribution Centers (94); Operations Research I (91); Operations Research II (92); Optimization Theory & Computational Methods (91); Probability Theory & Mathematical Statistics (91)

PUBLICATIONS (Corresponding Author*)

- [1] **Yijian Lu**; Yining Yu. 2026 "Two-Stage Stochastic MPEC Framework with Soft Time Windows for Multimodal Container Network Design: A Dynamic Benders Decomposition with Adaptive Cut Scheduling" *Transportation Research Part E: Logistics and Transportation Review* [SCI Q1, IF:8.5] . *Under Revision*.
- [2] **Yijian Lu**; Feifei Li; Yining Yu*. 2026 "Carbon-Aware Multi-Project Scheduling in Prefabricated Construction: A Makespan-Carbon Trade-off Model and an Interpretable Learning-Guided Matheuristic" *Computers & Operations Research* [SCI Q1, IF:4.6] . *Under Review*.
- [3] **Yijian Lu**; Menghao Li; Yiheng Fu; Ge Wu; Yining Yu*. 2026 "Emergency transfer of stranded people by synchronised heavy-lift drone coalitions: Routing, scheduling, and conditional exact completion" *Transportation Research Part C: Emerging Technologies* [SCI Q1, IF:8.4] . *Under Review*.
- [4] **Yijian Lu**; Yizhi Zhen; Fangni Zhang. 2026 "Cooperative Bus-UAV Last-Mile Delivery with Station Lockers and Fixed-Timetable Synchronization: A Mixed Integer Linear Program Model and Valid Inequalities" *Transportation Research Part C: Emerging Technologies* [SCI Q1, IF:8.4] . *Under Review*.
- [5] Zihan Zhang; Jiani Li; Haoyu Yang; Zengji Zheng*; **Yijian Lu***. 2026 "A Tolerance-Aware Pricing Certificate for Stabilized Dantzig-Wolfe Branch-and-Price in Unit Commitment" *Algorithms* [EI Q2, IF:2.6] . *Under Revision*.
- [6] Yining Yu; **Yijian Lu***. 2025 "Exploiting Structural Properties of the Allocation Subproblem in Emergency Logistics Center Location: A Dominance-Preserving Decoder and Heuristic-Priority Discrete PSO" *Computers & Industrial Engineering* [SCI Q1, IF:7.3] . *Under Revision*.

SELECTED RESEARCH EXPERIENCE

Carbon-Aware Multi-Project Scheduling in Prefabricated Construction: A Makespan-Carbon Trade-off Model and an Interpretable Learning-Guided Matheuristic

Primary Researcher

Sep 2025 - May 2026

- Optimised enterprise-level multi-project scheduling for prefabricated construction under dual-carbon mandates, incorporating GB/T 51366 staged carbon emissions across production-transport-installation chains, project due dates with weighted tardiness, on-site inventory buffers, and time-varying renewable resources.
- Formulated a bi-objective makespan-carbon MILP model solved via ϵ -constraint, and designed an Interpretable Learning-Guided Cooperative Matheuristic (LM-CMA) combining objective-aligned LP-relaxation priority learning, two-layer co-evolution, bottleneck fix-and-optimize re-optimization, and forward-backward improvement.
- *First-authored* manuscript submitted to *Computers & Operations Research*; achieved an overall mean gap of **21.1%** outperforming *GA (32.1%)*, *PSO (54.8%)*, and priority rules (86.0%) across 36 benchmark instances, and identified a critical carbon cap where tardiness costs jump up to 15-fold when tightened below the threshold.

Emergency transfer of stranded people by heavy-lift drones: Routing and scheduling under operational admissibility screens

Primary Researcher

Jul 2025 - Jun 2026

- Optimised tactical emergency evacuation of stranded disaster victims using synchronised heavy-lift drone coalitions, accounting for receiving site capacities, individual aircraft energy budgets, three outcome states (on-time, late, unserved), and non-reusable terminal recovery slots.
- Formulated a 3-level lexicographic MILP model, proved strong NP-hardness alongside polynomial exact completion for fixed profiles, and developed Adaptive Conditional-Profile Evaluation Search (Adaptive CPES) embedding exact profile completion, Q-learning ALNS, and tied-critical-DAG minimum-cut rebuilds.
- *First-authored* manuscript submitted to *Transportation Research Part C: Emerging Technologies*; achieved the smallest mean empirical shortfall (10.76%) among standalone heuristics on 54 benchmark instances, and **reduced pooled shortfall from 7.34% to 0.59%** when serving as a primal start generator for exact MILP solving.

A Tolerance-Aware Pricing Certificate for Stabilized Dantzig-Wolfe Branch-and-Price in Unit Commitment

Primary Researcher

Nov 2025 - Jul 2026

- Developed a conditional, tolerance-adjusted node-pricing certificate for stabilized Dantzig-Wolfe (DW) branch-and-price in deterministic unit commitment, bridging the gap between exact subproblem workload reduction and rigorous

- branch-and-price node termination certification.
- Designed CSDW-S featuring a Phase-II lower-bound cascade with period-separable operational-envelope bounds (Tier E), chronology-preserving ramp-relaxed dynamic programming (Tier D), and exact pricing-MIP fallbacks under master-error budgets and one-sided rounding.
- Corresponding-authored** manuscript submitted to Algorithms; achieved an 80.9% median reduction in exact unit pricing solves across 20 certified root cells and a 75.0% workload reduction in exact true-dual unit solves (7,409 vs. 19,908) while maintaining mathematically rigorous certified lower bounds.

Cooperative Bus-UAV Last-Mile Delivery with Station Lockers and Fixed-Timetable Synchronization: A Mixed Integer Linear Program Model and Valid Inequalities

Primary Researcher

Sep 2025 - May 2026

- Optimised order assignment and outbound routing for a public-bus-UAV cooperative last-mile delivery system along Hong Kong transit corridors, considering bus timetables, locker capacity and UAV energy constraints.
- Developed a Corridor-State Compressed Branch-and-Cut (CSBC) algorithm by exploiting corridor-specific structural properties; validated the model on 28 GTFS-based benchmark instances using real KMB bus routes and timetables.
- First-authored** manuscript submitted to Transportation Research Part C: Emerging Technologies; achieved **3.1×faster** optimality proofs, **solved 6/12 medium-scale instances** where baseline methods failed, and **improved large-scale incumbent solution quality by 12-55%**.

Exploiting Structural Properties of the Allocation Subproblem in Emergency Logistics Center Location: A Dominance-Preserving Decoder and Heuristic-Priority Discrete PSO

Primary Researcher

May 2025 - Nov 2025

- Optimised emergency logistics centre location for Zhengzhou flood response, minimising aggregate transportation, set-up and operations costs under dynamic demand and time-critical delivery constraints.
- Designed a dominance-preserving decoder with two heuristic-priority rules embedded in a discrete PSO framework, validated on large-scale benchmarks and Zhengzhou flood data with multi-parametric sensitivity analysis.
- Corresponding-authored manuscript submitted to Computers & Industrial Engineering; achieved **7.3% cost reduction** over exact methods, **52% faster** convergence than baseline PSO, and **held cost deviation below 5%** under ±20% demand shocks.

Two-Stage Stochastic MPEC Framework with Soft Time Windows for Multimodal Container Network Design: A Dynamic Benders Decomposition with Adaptive Cut Scheduling

Primary Researcher

Oct 2024 - Apr 2025

- Designed a multimodal container network for the China-ASEAN Sea-Rail Corridor, minimising strategic hub investment and expected recourse cost under stochastic OD demand, soft time-window penalties and lower-level user equilibrium.
- Formulated a Two-Stage Stochastic MPEC with linearised KKT conditions and developed a Dynamic Benders Decomposition with Adaptive Cut Scheduling (DBD-ACS) using a novel z-relaxation bridge theorem to recover Pareto-optimal cuts.
- First-authored** manuscript submitted to Transportation Research Part E: Logistics and Transportation Review; solved all 18 benchmarks to proven optimality with a **23.57×stronger lower bound**, cutting the real-world terminal gap **from 96.87% to 26.28% within 1 hour**.

SF Cold Chain Logistics Routing Optimization

Key Member

Jun 2024 - Oct 2024

- Optimised multi-temperature-zone vehicle routing for SF Express, jointly handling frozen, chilled and ambient fruit deliveries under hard time windows and insulation-board capacity limits.
- Formulated a mixed-integer programme with partition-based temperature zoning, time-of-day differentiated cost and quadratic early-arrival penalty; solved exactly via CPLEX and benchmarked against a team-developed LNS heuristic.
- Co-authored technical report won the **Outstanding Proposal Award from SF Logistics**; the new routing plan delivered a **12.7% total-cost reduction** over historical empirical routes.

Cold Chain Hub Location Optimization for Kiwifruit under Rural Revitalization

Key Researcher

Nov 2023 - Jan 2024

- Solved a stochastic facility-location problem for kiwifruit distribution across 7 townships under yield uncertainty, minimising construction, operations, transportation and wastage costs across a farm-pre-cooling-sales multi-echelon network.
- Built a two-stage stochastic programme with 12 demand scenarios and implemented a Benders decomposition (binary location master / dualised linear subproblem) in MATLAB.
- Won the **Outstanding Paper Award** at the University Innovation Competition; the optimised hub plan achieved a **26% total-cost reduction** compared with the historical layout.

LEADERSHIP, VOLUNTEERING & INDUSTRY EXPERIENCE

SF Express (Shenzhen) Co., Ltd. Shenzhen, China

Headquarters Management Trainee

Jun 2024 - Oct 2024

- Rotated through frontline pickup, delivery and warehousing operations; personally handled **1,000+** parcels across urban villages, CBD towers and university campuses, mastering the full courier daily SOP and last-mile constraints.
- Re-engineered warehouse sortation and station-level parcel allocation SOP by re-sequencing inbound-outbound flows and rebalancing courier zones, lifting peak-hour sortation throughput by **18%** and cutting average loading time per route by **12%**.

College Student Science and Technology Innovation Association of NJAU

President

Mar 2023 - Jan 2024

- Led **10+** innovation and business simulation events, engaging over **500** student participants and coordinating teams, judges and event delivery across applied technology, logistics and entrepreneurship activities.

Nanjing Jinling Library & Campus Reading Initiatives

Volunteer

Sep 2022 - Jan 2025

- Supported reading literacy, library cultural promotion and academic tutoring initiatives, engaging over **300** primary, secondary and orphanage students.
- Awarded Outstanding Reading Recommendation Officer by Nanjing Agricultural University.

HONORS & AWARDS

SCHOLARSHIP

- [1] First-Class Scholarship (**Top 5%**), Nanjing Agricultural University | 2024
- [2] Second-Class Scholarship (Top 15%), Nanjing Agricultural University | 2023
- [3] Individual Scholarship for Sci-tech Innovation(**Top 5%**), Nanjing Agricultural University | 2023
- [4] Individual Scholarship for Sports(**Top 5%**), Nanjing Agricultural University | 2023
- [5] Individual Scholarship for Social Practice(**Top 5%**), Nanjing Agricultural University | 2023

AWARDS

- [1] **Gold Award**, College Students Innovation and Entrepreneurship Training Program (Provincial Level) | 2024
- [2] **Bronze Award**, College Students Innovation and Entrepreneurship Training Program (Provincial Level) | 2024
- [3] **First Prize**, The Challenge Cup, College of Information Management | 2024
- [4] **Outstanding Student**, Summer Camp of the University of Aberdeen | 2023
- [5] **Second Prize**, Provincial Environmental Knowledge Competition for College Students | 2023
- [6] **First Prize**, Men's 100m Breaststroke, The 51st Sports Competition of Nanjing Agricultural University | 2023
- [7] 3rd Place, Men's 4x100m Medley Relay, The 20th Jiangsu Games (Jiangsu Provincial Sports Games) | 2022
- [8] **Second Prize**, Provincial Enterprise Competition Simulation Contest | 2022
- [9] Third Prize, Provincial Enterprise Competition Simulation Contest | 2021

SKILLS

- Languages: Chinese (Native), English (Full Professional Proficiency; **Master's degree fully completed in English**).
- Software: MATLAB, PYTHON, CPLEX, GUROBI, SPSS, SQL SERVER, LINGO etc.
- Sports: Badminton, Fitness Training, Competitive Swimming, Vice President of Badminton Association in NJAU, **Five-level of Social Sports Instructors (National Vocational Qualification Certificate)**.